

Ultrametric Quantum Error Correction

Testing Discrete Geometry in the Redundancy–Fidelity Tradeoff

CWI Summer School on Quantum Algorithms & Quantum Error Correction 24–28 August 2026
Poster Session: Wed 26 Aug, 16:30–18:00 · Poster Award Ceremony: Thu 27 Aug, 08:50

Panel 1 — The Hypothesis

- Standard view:** QEC assumes noise accumulates over a continuous topology; logical fidelity improves smoothly as redundancy is added.
- Alternative to test:** error propagation follows a discrete, hierarchical geometry — a Bruhat–Tits tree T_p — and redundancy quantizes in units of $(p+1)$.
- Signature to look for:** a staircase — flat plateaus followed by sudden jumps — instead of a smooth curve.

STATUS: HYPOTHESIS ONLY — NO EXPERIMENTAL RESULT CLAIMED

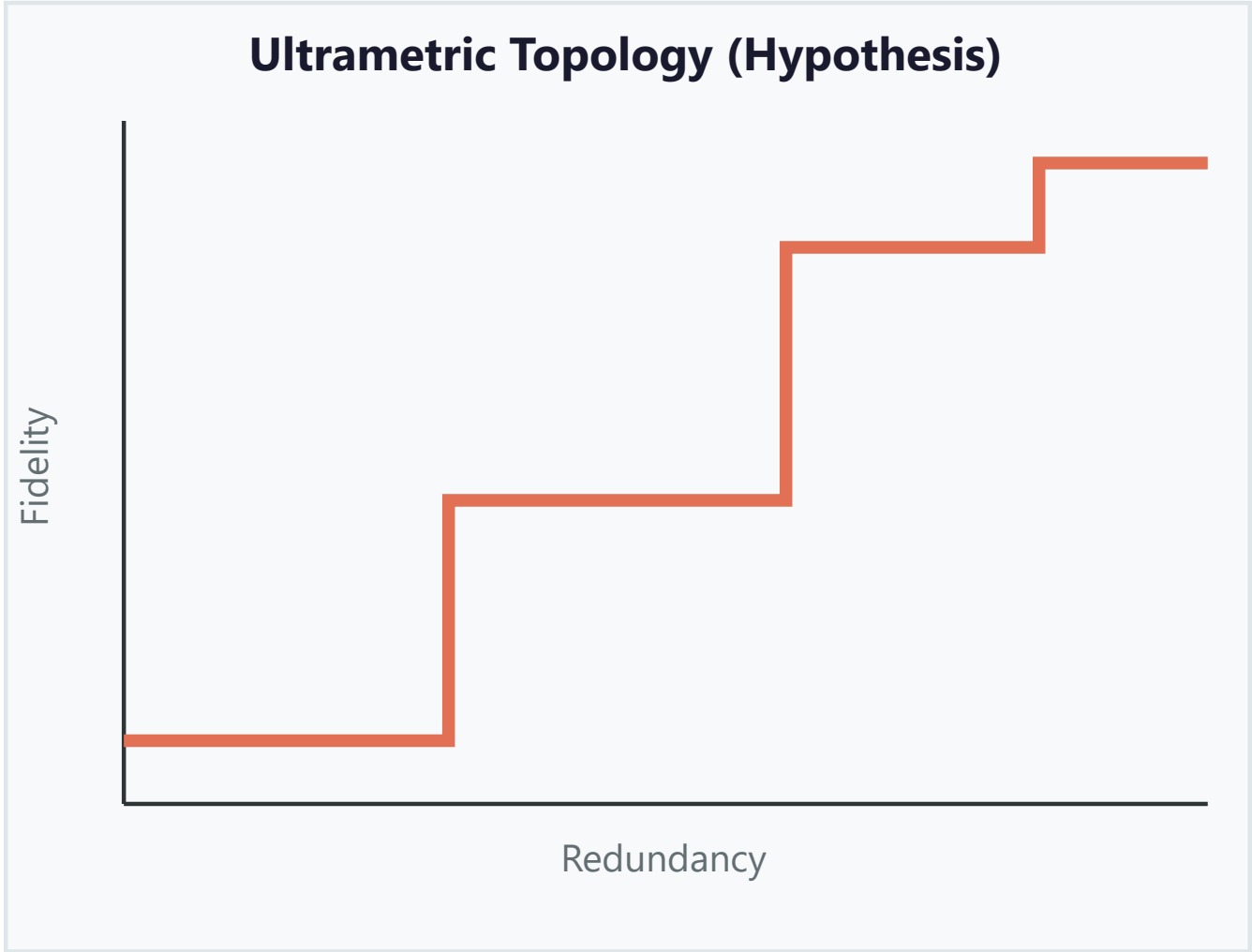
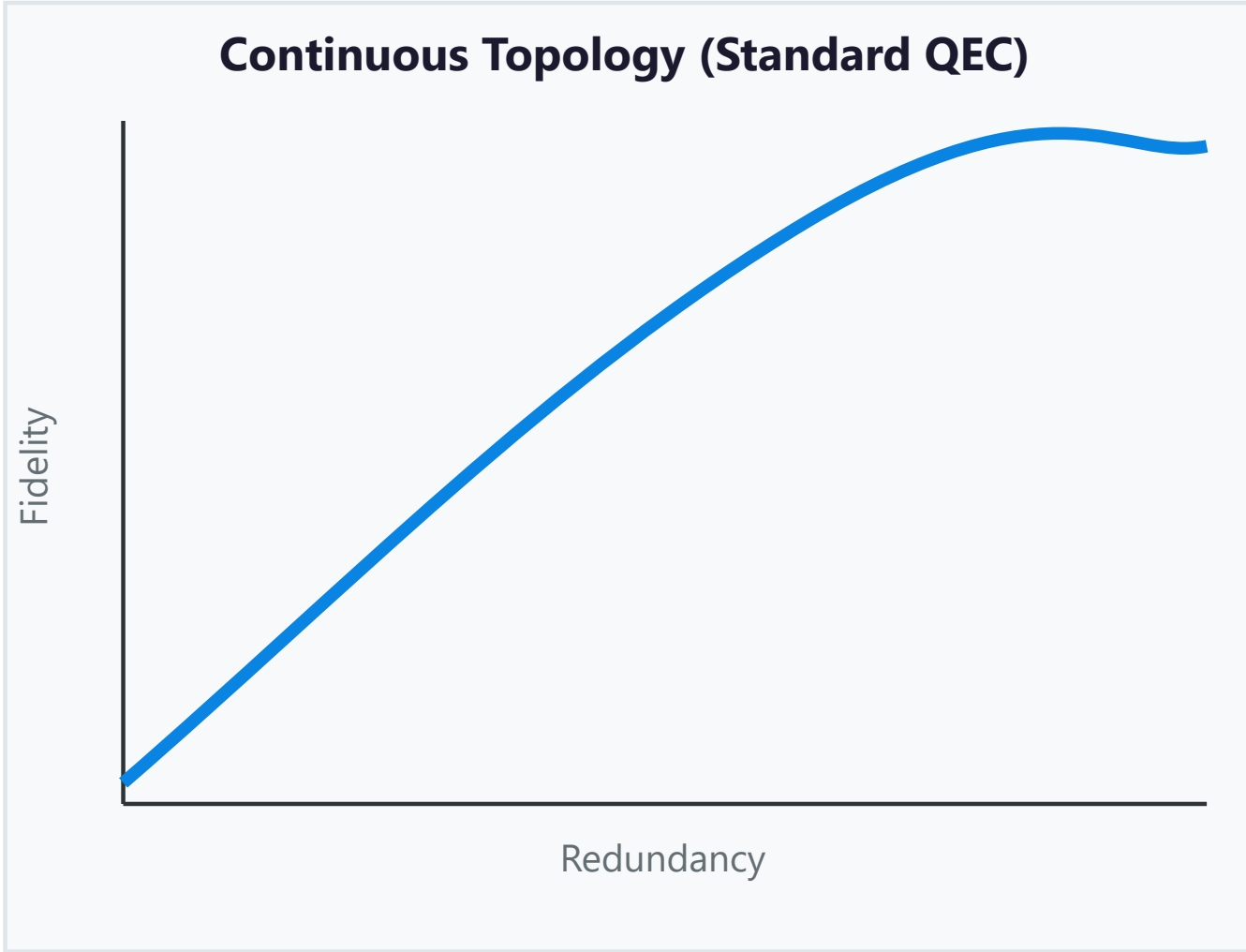
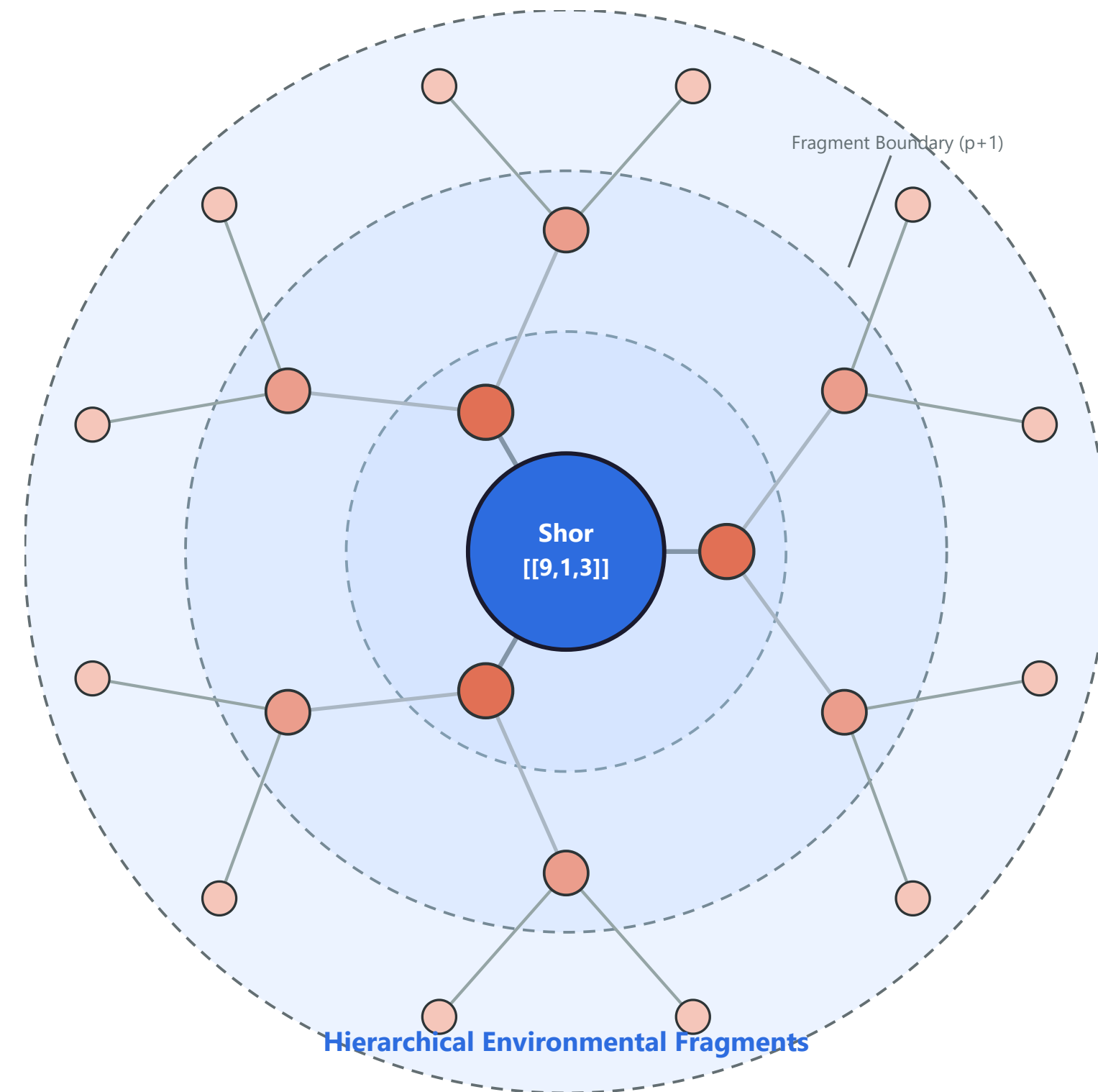


Figure 1: smooth fidelity curve (continuous topology, left) vs. staircase (ultrametric hypothesis, right). Hypothesis only; not observed.

Panel 2 — A Proposed Experiment (not yet run)



- Encode:** one logical qubit in the Shor $[[9,1,3]]$ code.
- Interact:** expose it to a fixed, structured environment of spectator qubits.
- Correct:** run the full error-correction circuit; measure final logical fidelity.
- Scale:** group environment qubits into larger hierarchical fragments; track fidelity across fragment boundaries.

Smooth shift with fragment size = continuous model. Staircase = ultrametric model.

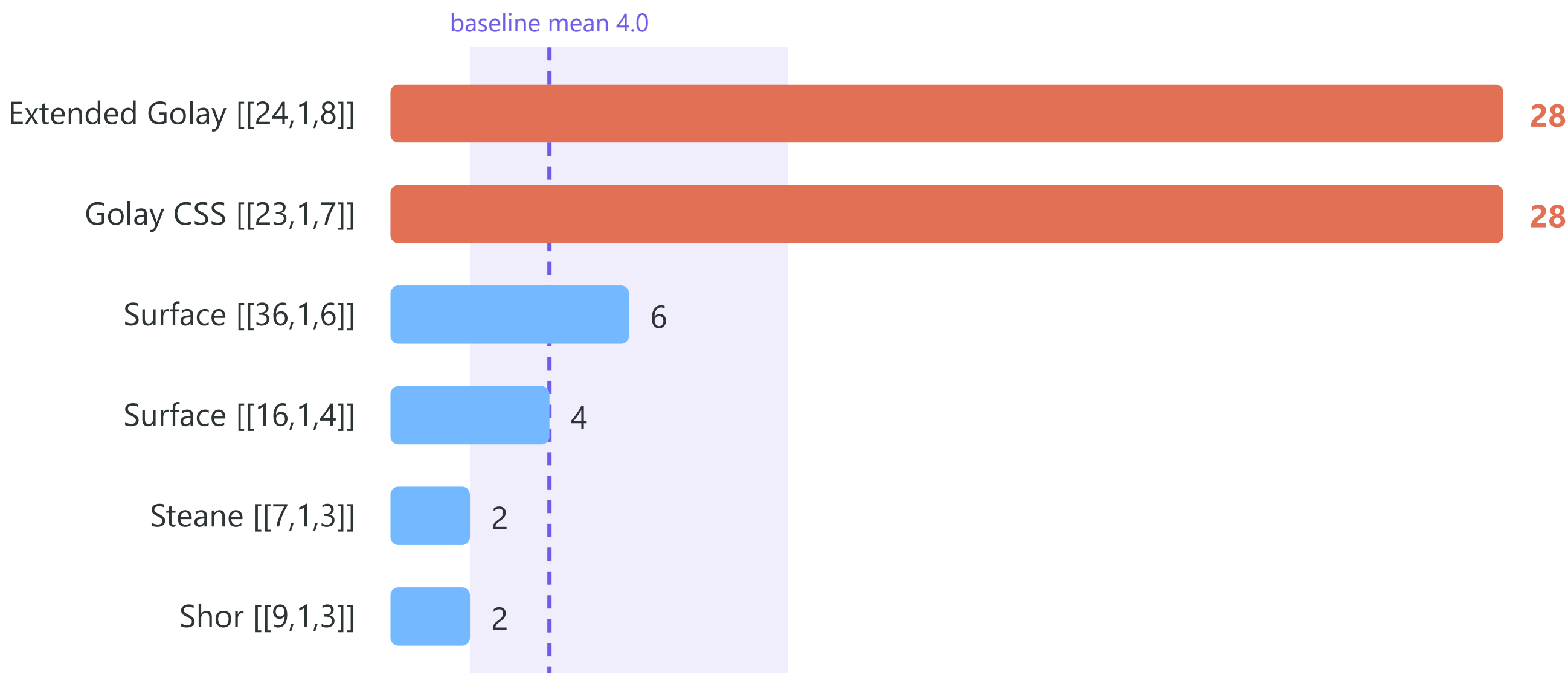
What exists today: a provisional numerical toy model (Archimedean Shadows) suggested redundancy may quantize in discrete steps (earlier versions reported 3, 6, 12, 24) — reported provisionally — and its own most seductive signature (the “forbidden window”) **failed its own falsification test**. Null reported, not hidden.

Panel 3 — The Cross-Check I Already Ran (it mostly failed — honestly)

Witness idea: 2-adic valuations as a structural fingerprint. Conjecture C7.3: the Mahler-spectrum valuation v_p^{max} separates “optimal” codes (28) from random codes ($=4$).

v_p^{max} by code family — only Golay-type self-dual codes separate

Random-ensemble baseline (100 trials, $n=16$, $k=4$): mean 4.0, range 2–10 · Source: ACRP-06, 10.5281/zenodo.21754148



Panel 4 — Value & Open Questions

A smooth curve validates continuous models; a staircase would force code architectures designed for hierarchical noise.

- What physical mechanism would produce tree-like error geometry?
- Does any valuation witness predict where the staircase should be most visible? (v_p^{max} does not.)
- How would discrete geometry alter asymptotic fault-tolerance thresholds?

I'm here to learn: has anyone in this room seen staircase-like scaling — or a reason it must be wrong?

Panel 5 — The Diagrammatic Bridge (ZX-calculus)

SPEAKING THE ROOM'S LANGUAGE — SAME MAP, DIFFERENT NOTATION

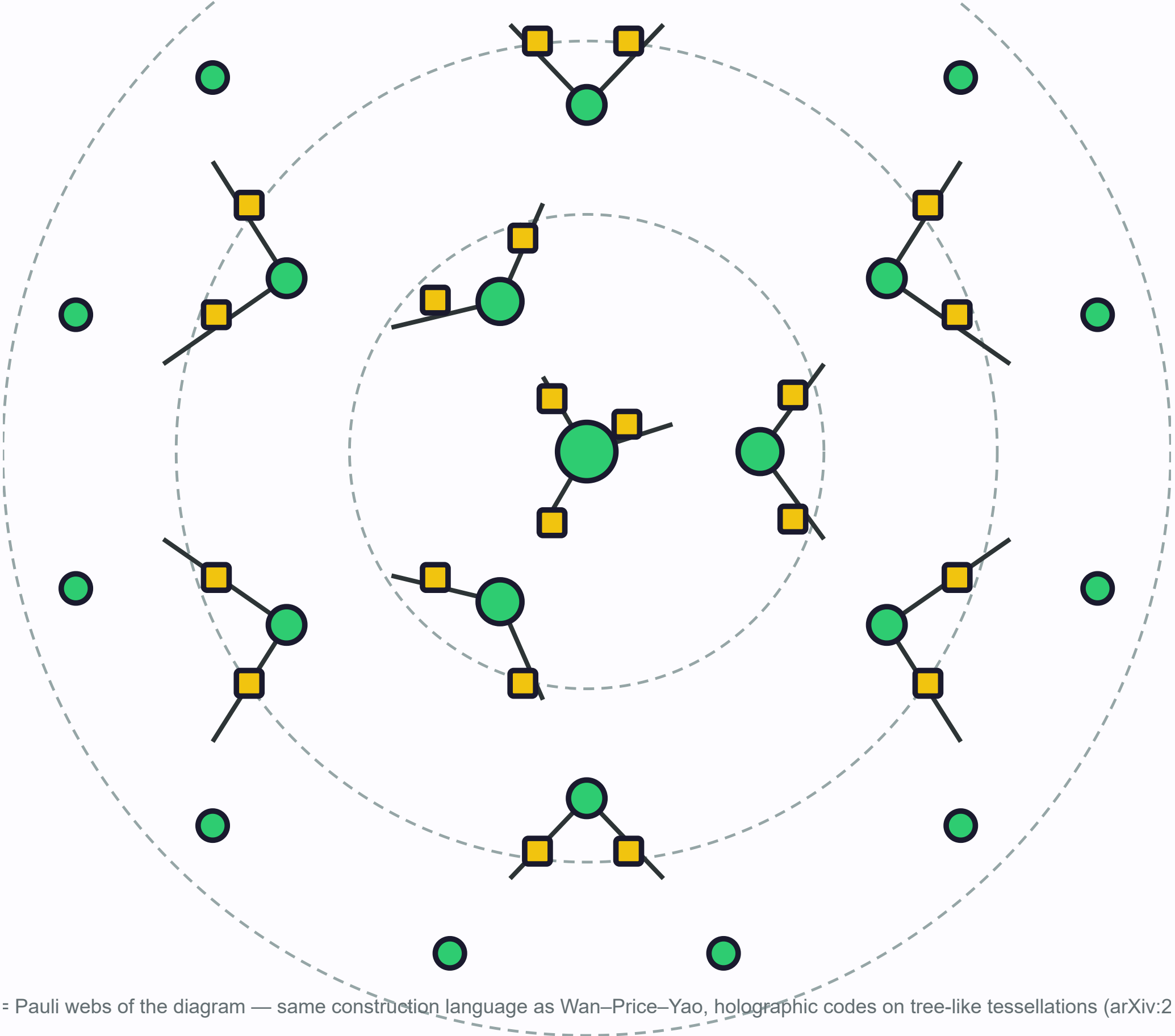
- Spiders — the atom.** Green = Z-basis copy/merge, red = X-basis. Graph-state convention: green spider (0 in, 1 out) = $|+\rangle$; yellow H-box (2 in, 2 out) = CZ = the graph-state edge.
- Pauli webs — the stabilizers, read off the diagram.** A connected sub-diagram that evaluates to a Pauli operator: the code's stabilizers and logical operators as paths through the spider web. This is the term the QEC-meets-ZX literature uses (Wan–Price–Yao, arXiv:2601.04467).
- Gadgets — the field's cost unit.** Phase/Pauli gadgets $e^{i\theta P}$. T-count optimization is gadgetization (Duncan–Kissinger–Perdrix–van de Wetering, arXiv:1902.03178; Pauli Gadget Synthesis, QPL 2026). The algebraic cousin of an energy cost — the seam this poster cares about.

The one-line statement: “I represent the ultrametric geometry as a ZX graph state — the same language as the holographic-code construction on tree-like tessellations (Wan–Price–Yao 2026). The stabilizers are the Pauli webs of the diagram; the hierarchy is the tree.”

The honest caption (map ≠ territory): the diagrams import structure from different silos — internally consistent, externally unprobed. This caption says so. Acknowledging the seam is the legitimacy move (APS §6.2, audit the map). “It's a map, *not* the territory — but it's the same map the field draws.”

Bruhat–Tits tree T_p (truncated, depth 3) as a ZX graph state

- green spider = $|+\rangle$ · yellow H-box = CZ edge · $p=2 \Rightarrow (p+1)=3$ -regular branching



Pauli webs of the diagram — same construction language as Wan–Price–Yao, holographic codes on tree-like tessellations (arXiv:2

Figure 4: the ultrametric geometry, drawn in the room's own diagrammatic language. Verified: 22 spiders = 22 vertices, 21 H-boxes = 21 edges.

References (Zenodo concept DOIs — verified 2026-08-16)

Ultrametric Code Spaces — 10.5281/zenodo.21624194
Archimedean Shadows (v1.1) — 10.5281/zenodo.21809888
Ultrametric Foundations (C7.3) — 10.5281/zenodo.21193003
ACRP-06 v_p^{max} extension — 10.5281/zenodo.21737221
Adelic Cross-Domain Program v5 — 10.5281/zenodo.21691414
“What Remains” — 10.5281/zenodo.21922812

ZX references (arXiv — verified 2026-08-18)

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v3 — ZX panel added 2026-08-18

